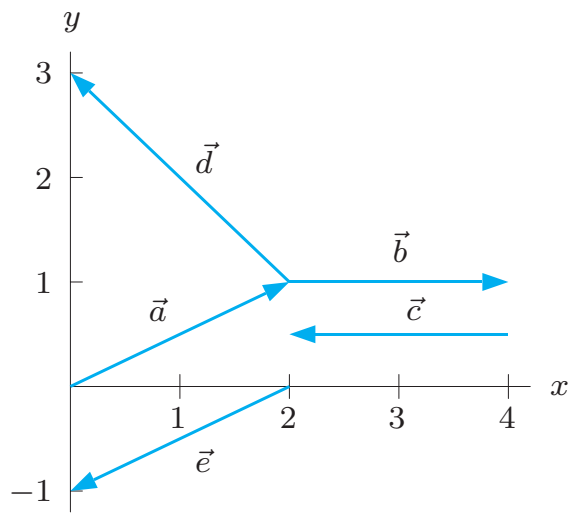
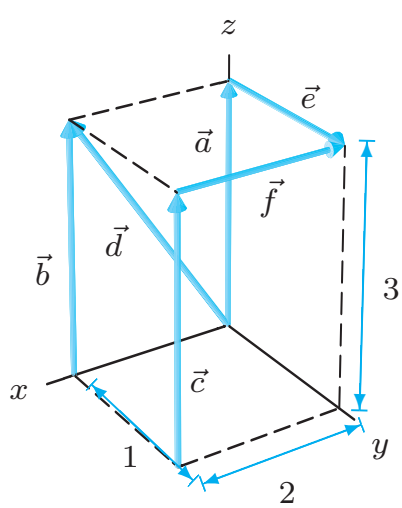
Suggested problems: 1-33 odd

Problem 1: Resolve the following vectors into the basis vectors.



$$\vec{a} = 2\vec{i} + \vec{j}$$
$$\vec{b} = 2\vec{i}$$
$$\vec{c} = -2\vec{i}$$
$$\vec{d} = -2\vec{i} + 2\vec{j}$$
$$\vec{e} = -2\vec{i} - \vec{j}$$

Problem 2: Resolve the following vectors into the basis vectors.



$$\vec{a} = 3\vec{k}$$
$$\vec{b} = 3\vec{k}$$
$$\vec{c} = 3\vec{k}$$
$$\vec{d} = 2\vec{i} + 3\vec{k}$$
$$\vec{e} = \vec{j}$$
$$\vec{f} = -2\vec{i}$$

Problem 3: Perform the following operations.

(a) $-4(\vec{i} - 2\vec{j}) + (-3)(2\vec{i} + \vec{j})$

$$-10\vec{i} + 5\vec{j}$$

(b) $\frac{1}{2}(2\vec{i} - \vec{j} + 3\vec{k}) + 3(\vec{i} - \frac{1}{6}\vec{j} + \frac{1}{2}\vec{k})$

$$4\vec{i} - \vec{j} + 3\vec{k}$$

Problem 4: Find a vector with length 2 which points in the same direction as $\vec{v} = \vec{i} - \vec{j} + 2\vec{k}$.

$$2 \cdot \frac{\vec{v}}{||\vec{v}||} = \frac{2}{\sqrt{1+1+4}}(\vec{i} - \vec{j} + 2\vec{k}) = \frac{2}{\sqrt{6}}\vec{i} - \frac{2}{\sqrt{6}}\vec{j} + \frac{4}{\sqrt{6}}\vec{k}$$

Problem 5: Find the values of a making $\vec{v} = 5a\vec{i} - 3\vec{j}$ parallel to $\vec{w} = a^2\vec{i} + 6\vec{j}$.

We need to find a number λ such that $\lambda\vec{v} = \vec{w}$. This would mean

$$5\lambda a\vec{i} - 3\lambda\vec{j} = \lambda a^2\vec{i} + 6\lambda\vec{j},$$

so that $5\lambda a = a^2$ and $-3\lambda = 6$. The second equation says that $\lambda = -2$, so the first equation becomes $-10a = a^2$. Solving for a gives $a(a + 10) = 0$, so that $a = 0, -10$.